**BRAINWARE UNIVERSITY**
**SCHOOL OF COMPUTATIONAL & APPLIED SCIENCES**
**DEPARTMENT OF COMPUTATIONAL SCIENCES**
**BACHELOR OF COMPUTER APPLICATIONS /**
**BACHELOR OF COMPUTER APPLICATIONS (HONOURS) /**
**BACHELOR OF COMPUTER APPLICATIONS (HONOURS with RESEARCH) - 2024**
**SEMESTER –V**

**Course Code: BCA57115**
**Course Name: Full-stack Development-II**
**Weekly Contact Hours: 2L+4P**
**Credits: 4**
**Total Allotted Hours: 30L+ 60P**

**Course Objective:** The aim of this course is to understand the different aspects of AngularJS and ReactJS. Different events, directives and the complete overview of the subject has been covered in this syllabus. The different services, events and deployment will be explored in details.

**Pre-requisite(s):** Basic knowledge of operating systems and java scripts.

**Course Outcome (COs):** After the completion of the course, students would be able to:

**CO1:** Understand and compare the fundamentals of ReactJS and its architecture.
**CO2:** Discuss and analyze the module, controllers and events.
**CO3:** Explain and apply react data binding.
**CO4:** Design and evaluate forms, modules with directives.
**CO5:** Build and evaluate services on different platforms.

**Module 1: React JS and JSX with Component** **[4H(L)]**

Introduction to React JS, prerequisites of react JS, why use React JS, software and description, install react JS, creating react app. converting HTML JSX, file and directory structure of react, code snippets, javascript variable in JSX, comments, class component, function component.

**Practical:** **[8H(P)]**

Installation of react JS, implementation of creating react app. converting HTML JSX, file and directory structure of react, code snippets, javascript variable in JSX, comments, class component, function component.

**Module 2: React CSS with Bootstrap package and router concepts** **[6H(L)]**

Inline styling, CSS style sheet, CSS module, styled components, adding bootstrap for react, react bootstrap package, react table installation and use. install a react router, add a router in react, concept of props, use of props, passing props to child, create reusable component, props validation, concept of state, state rules, use of state.

**Practical:** **[12H(P)]**

Implementation of react CSS with Bootstrap and router.

**Module 3: User Input Forms with Events and deployment** **[6H(L)]**

React arrow function, handling forms, creating event handlers, free react developer tools, deploy react app on server.

**Practical: [12H(P)]**

Implementation of react arrow function, handling forms, creating event handlers, free react developer tools, deploy react app on server.

**Module 4: API Request with React and data binding [6H(L)]**

Fetching data, send data, API create using PHP & MYSQL, introduction to angularJS, MVC architecture, conceptual overview, setting up the environment, first application understanding and attributes, number and string expressions, object binding and expressions, working with arrays forgiving behavior, understanding data binding.

**Practical: [12H(P)]**

Implementation of API and related applications.

**Module 5: Working of directives using Controllers with filters and services [8H(L)]**

Conditional directives, understanding controllers programming controllers & $scope object, adding behavior to a scope object with parameters, nested controllers and scope inheritance, multiple controllers and their scopes, built-in filters, creating custom filter, using simple form, using CSS classes, form events, custom validations, understanding services, developing creating services, using a service, injecting dependencies in a service.

**Practical: [16H(P)]**

Implementation of scope inheritance using controller and use of validation control.

**Course Outcomes (COs) and Modules Mapping:**

| Sl. No. | Course outcome (COs) | Mapped Module(s) | Learning levels (as per Bloom's Taxonomy from K1 to K6) |
|---|---|---|---|
| 1. | CO1 | M1 | K2, K5 |
| 2. | CO2 | M2 | K2, K4 |
| 3. | CO3 | M3, M4 | K2, K3. K5 |
| 4. | CO4 | M4 | K5, K6 |
| 5. | CO5 | M5 | K5, K6 |

**Text books:**

1. Learning React: Functional Web Development with React and Redux, Banks, Alex, Porcello, Eve, O'Reilly Media, Inc., 1st Edition, 2017.
2. Complete Book on Angular 4,Nate Murray, Felipe Coury, Ari Lerner and Carlos Taborda , CreateSpace Independent Publis , 1st Edition, 2017.

**Reference books:**

1. The Complete Book on Angular js, AriLerner, Ng-Book, Lightning Source Inc;1st edition, 2010.